

End of Day Report

Name	Sumukh Naidu
Designation	AI Engineer
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Summary

Closed out remaining page-selection bugs in Crawl, revisited the BRD directly to clarify tier structure and priorities, did short R&D on small reasoning models for the upcoming Tier 1 escalation, and built the full CWI-09 benchmark and evaluation harness — a standalone package that hand-labels ground truth per company and scores the real extractor against it, replacing manual eyeballing of terminal output with actual precision/coverage numbers. All 43 unit tests for the new harness pass.

Key Activities

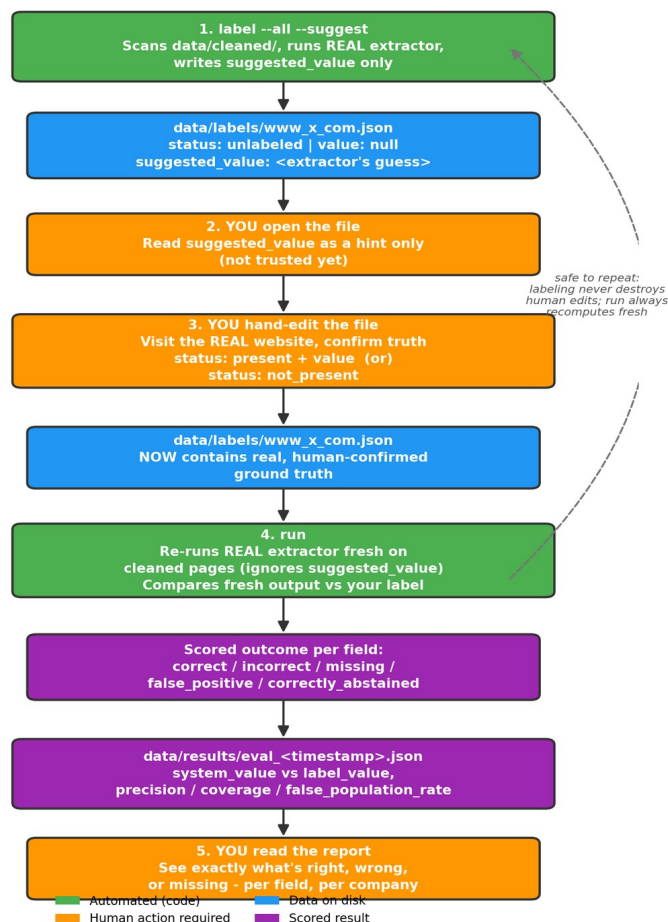
1. Fixed a second round of page-prioritization bugs in crawler.py: after removing the overly generic "company" keyword (which matched unrelated words inside article slugs), whole-segment matching then caused a false negative, missing a real /aboutus page on impactodigifin.com. Replaced both approaches with an explicit KEYWORD_VARIANTS whitelist (about/aboutus, contact/contactus, team singular only, etc.), verified against both the original false-positive case (wipro.com case studies) and the new false-negative case.
2. Reviewed the actual uploaded BRD document directly rather than relying on secondhand summary, and confirmed two distinct tier systems exist: Field tiers (Tier 1 core / Tier 2 valuable / Tier 3 opportunistic — which fields matter) and the Model Escalation Ladder (Tier 0 deterministic / Tier 1 small LM / Tier 2 mid LM / Tier 3 large-hosted — which method extracts a field). Confirmed the project is currently Tier 0-only within the escalation ladder.
3. Did R&D on small reasoning models for the upcoming Tier 1 escalation (per the BRD's own representative-model list), evaluating current options against the project's actual constraints (per-field call volume, citation-span grounding, cost/latency) rather than generic benchmark rankings; SmoLLM3-3B identified as the leading candidate given its dual-mode reasoning toggle, with Phi-4-mini-reasoning as a strong second option for numeric fields.
4. Identified a real gap against BRD Section 8.1: confidence scores in extractor.py are currently hand-picked constants, not the four-signal composition (source authority, corroboration, tier used, validation outcome) the BRD specifies, nor calibrated against real outcomes. Scoped this properly as belonging to the future Resolve stage rather than patching it into Extract.
5. Reviewed the BRD's explicit sequencing requirement: the benchmark/eval harness (CWI-09) must be completed before extraction work begins, and citation verification (CWI-10) is called "the single most important requirement" in the document. Decided to build CWI-09 first, since Tier 0's deterministic methods can't hallucinate the way an LLM would — citation verification's urgency is tied to the future Tier 1 small-model escalation, while a working benchmark gives real, measurable numbers immediately.
6. Designed and implemented the full CWI-09 benchmark/eval harness as a new top-level benchmark/ package, kept structurally separate from app/ so evaluation-only code (e.g. a naive placeholder merge standing in for the not-yet-built Resolve stage) can never be imported by production pipeline code.
7. The harness reads only already-cleaned data (data/cleaned/*.json) and makes zero network calls. Two commands: label (generates/refreshes hand-label templates per company, optionally pre-filled with the real

extractor's current output via `--suggest`, without ever overwriting confirmed human entries) and `run` (re-runs the real extractor fresh, scores it against hand-confirmed labels, and writes a timestamped report).

8. Scoring classifies every field into one of six outcomes (`correct`, `incorrect`, `missing`, `false_positive`, `correctly_abstained`, `unlabeled`) and computes precision, coverage, and `false_population_rate` — reporting null rather than a fake 0.0 when a denominator is zero. Citation validity is reported as an explicit not-yet-measured sentinel rather than a faked number, pending CWI-10.

9. Wrote and ran the full test suite (`test_labels`, `test_merge`, `test_normalize`, `test_scorer`) — 43 tests, all passing — covering idempotent/non-destructive labeling, all six scoring outcomes, per-field match-type rules (`exact`, `fuzzy`, `phone`, `URL`), and the naive merge's tie-break behavior.

How the Benchmark Harness Works



Running `label --all --suggest` scans already-cleaned pages and writes a template per company, pre-filled with the real extractor's current guesses under `suggested_value` only. A human then visits the actual website, confirms the true value, and hand-edits status and value directly (or marks a field `not_present` if genuinely absent) — this step is the only place ground truth enters the system. Running `run` re-executes the real extractor fresh against the cleaned pages, ignoring `suggested_value` entirely, and scores the fresh output against the hand-confirmed label, producing a timestamped report with `system_value` shown next to `label_value` for every field. Labeling is safe to repeat (never overwrites human edits), and every run recomputes from scratch, making this a genuine regression check as `extractor.py` evolves.

Upcoming Work

1. Hand-label real ground truth for the companies already crawled this week (Apple, Wipro, Microsoft, impactodigifin, etc.) and run the harness for the first real precision/coverage numbers on Tier 0 extraction.
2. Use those numbers to decide where Tier 0 genuinely needs improvement before considering any model escalation.
3. Build citation verification (CWI-10) once Tier 1 (small LM) work begins, using the harness's pending sentinel as the place its results will land.

4. Continue toward the Resolve stage — real confidence composition (source authority, corroboration, tier used, validation outcome) and conflict resolution, replacing both the extractor's placeholder confidence values and the harness's naive eval-only merge.